

PHY 115L
Introduction to the
Discrete Fourier Transform

Michael Mulhearn

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Chapter 2

Discrete Fourier Transform

2.1 Fourier Series for Complex-Valued Functions

We will treat the Fourier Series in the context of the inner product space of square-integrable periodic functions of period a , with an inner product defined as:

$$\langle f|g \rangle := \int_{-\frac{a}{2}}^{\frac{a}{2}} f^*(x)g(x)dx$$

The complex exponential functions

$$e_k(x) := \sqrt{\frac{1}{a}} \exp\left(i \frac{2\pi kx}{a}\right)$$

are orthonormal vectors. To verify:

$$\begin{aligned} \langle e_j|e_k \rangle &= \frac{1}{a} \int_{-\frac{a}{2}}^{\frac{a}{2}} \exp\left(i \frac{2\pi(k-j)x}{a}\right) dx \\ &= \frac{1}{\pi(k-j)} \cdot \frac{\exp(i\pi(k-j)) - \exp(-i\pi(k-j))}{2i} \\ &= \text{sinc}(\pi(k-j)) \end{aligned}$$

As k and j are integers, and sinc is zero for every integer multiple of π except for $\text{sinc}(0) = 1$, we have shown:

$$\langle e_j|e_k \rangle = \delta_{kj} \tag{2.1}$$

We will not derive it here, but the complex functions are also a complete basis, so we can expand any function in terms of the basis of complex exponentials by:

$$\psi(x) = \sum_{j=-\infty}^{+\infty} c_j e_j(x) \tag{2.2}$$

Because of orthonormality, we can determine the Fourier coefficients by:

$$c_k = \langle e_k|\psi \rangle \tag{2.3}$$

2.2 Discrete Fourier Transform

Next we want to consider the case where we only know the value of $\psi(x)$ at n discrete positions of x , specifically:

$$x_j = x_0 + \lambda j$$

for $x_0 = -a/2$, $j = 0, 1, \dots, n-1$ and λ the interval between data points:

$$\lambda = \frac{a}{n}$$

Notice that the sequence ends before $x = a/2$ because this point is redundant to $x = -a/2$ for the periodic functions that we are considering.

We account for the discrete sampling by making the replacement:

$$\psi(x) \rightarrow \sum_{m=0}^{n-1} \delta(x - x_j) \lambda f(x)$$

which is chosen because:

$$\int_{x_1}^{x_2} \sum_{m=0}^{n-1} \delta(x - x_j) \lambda f(x) dx = \sum_{m=m_1}^{m_2} \lambda f_m \xrightarrow{\lambda \rightarrow 0} \int_{x_1}^{x_2} f(x) dx$$

showing that integration (smoothing) of discrete sampled function reproduces the integral of the original function as the space between samples approaches zero. We do expect this sampling to introduce high-frequency artifacts which must be considered, which we will discuss later.

Calculating the fourier series coefficients with the sampled function we obtain:

$$\begin{aligned} c_k &= \langle e_k | \psi \rangle \\ &= \sqrt{\frac{1}{a}} \int_{-\frac{a}{2}}^{\frac{a}{2}} \exp\left(-i \frac{2\pi k x}{a}\right) \sum_{m=0}^{n-1} \delta(x - x_j) \lambda f(x) dx \\ &= \sqrt{\frac{\lambda}{n}} \sum_{m=0}^{n-1} \exp\left(-i \frac{2\pi m k}{n}\right) \exp\left(-i \frac{2\pi k x_0}{a}\right) f_j dx \end{aligned}$$

which we can reorganize as:

$$\frac{c_k}{\sqrt{\lambda}} \exp\left(i \frac{2\pi k x_0}{a}\right) = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} \exp\left(-i \frac{2\pi m k}{n}\right) f_m \equiv A_k \quad (2.4)$$

where A_k are the Fourier coefficients from `numpy.fft.fft` if called with the normalization option `norm='ortho'`. The complex exponential on the left hand side is a phase factor that accounts the location of the first data point x_0 . Notice that the right hand side depends only on indices, not the actual x positions.

Something very interesting has happened to our Fourier coefficients as a direct result of discretizing the data. Notice that:

$$c_{k+n} = c_k$$

so evidently we have only n unique Fourier coefficients.

Blah Blah Blah... Nyquist... Blah Blah blah.

$$\begin{aligned}
 \psi(x) &= \sum_{j=j_a}^{j_b} c_j e_j(x) \\
 &= \frac{1}{\sqrt{a}} \sum_{j=j_a}^{j_b} c_j \exp\left(i \frac{2\pi j x}{a}\right) \\
 &= \frac{1}{\sqrt{n}} \sum_{j=j_a}^{j_b} A_j \exp\left(i \frac{2\pi j (x - x_0)}{a}\right)
 \end{aligned}$$

2.3 Homework Problems

Problem 1: Use python `fft.fft` to calculate FS for a sine and cosine. Plot the magnitude and phase at each wave number and compare to what you expect.

Problem 2: Plot $f(x)$ by expanding using the Fourier coefficients from Problem 1. Do not use inverse discrete fourier transform! We want a smooth function!

Problem 3: For square wave, plot coefficients, then plot $f(x)$ obtained from expansion in fourier coefficients/

...

Problem 2: Write a program to determine the coefficients of the Hermite polynomial $H_n(u)$ for any degree n by using the generating function:

$$\exp(-z^2 + 2zu) = \sum_{n=0}^{\infty} \frac{z^n}{n!} H_n(u)$$

Print the non-zero coefficients for $n = 0$ to 5 and compare to e.g. Griffiths Table 2.1

Problem 3: (A) Write a program to calculate the Hermite polynomial $H_n(u)$ for a particular value u and for any degree n by using the recursion formula:

$$H_{n+1}(u) = 2uH_n(u) - 2nH_{n-1}(u)$$

.

(B) Test your function **thoroughly** against the Python library results (`scipy.special.hermite`) using randomly chosen values. That is, pick a random integer n from 0 to some reasonable maximum test value N , then pick a randomly chosen u value from some reasonably range, and compare the results of your program with that of the python library. You want to do this many many times, so you'll have to figure out a way to handle the fact the agreement will not be perfect and yet still say something definitive.

Problem 4: Plot the wave function $\psi_n(x)$ and $|\psi_n(x)|^2$ for $n = 0, 1, 2, 3$. You are aiming to reproduce Fig 2.7(a) in Griffiths. (Let me know if you don't have access to this textbook and I will include photocopies of the needed figures on the course site.) Remember that $\psi_n(x)$ is not just the Hermite polynomial! You may use the python library for the Hermite polynomials.

Problem 5: Numerically verify that the $\psi_n(x)$ are orthonormal. Keep to $n < 10$. (Depending on your approach, you won't get exactly 1 and 0 so you will need some way to handle this and still give a definitive answer...there are many valid approaches)